



Department
for Environment,
Food & Rural Affairs

Guidance

Recyclability assessment methodology: stages of recyclability and definitions

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Background

This Recyclability Assessment Methodology (RAM) is published by PackUK, the Scheme Administrator for Extended Producer Responsibility (EPR) for packaging. This version of the RAM is for use in assessing packaging material reported as having been supplied in 2025. PackUK will update this guidance annually in the Autumn in readiness for the following year's reporting.

From 1 January 2025, liable producers who supply household packaging must assess the recyclability of that packaging and report the results of the assessment to regulators (Environment Agency, the Scottish Environment Protection Agency, Natural Resources Wales or the Northern Ireland Environment Agency, as appropriate).

This is RAM version 1.1 (April 2025) which is a simplified version 1.0 published on 23 December 2024, following feedback from the packaging value chain. Any significant changes made have been highlighted at the end of each section, any minor changes have been made within the body of the text.

You should read this guide alongside other RAM information that covers:

- [how to assess your packaging waste \(https://www.gov.uk/guidance/recycling-assessment-methodology-how-to-assess-your-packaging-waste\)](https://www.gov.uk/guidance/recycling-assessment-methodology-how-to-assess-your-packaging-waste)
- [RAM ratings and outputs for different packaging materials \(https://www.gov.uk/government/publications/recycling-assessment-methodology-materials-and-outputs\)](https://www.gov.uk/government/publications/recycling-assessment-methodology-materials-and-outputs)
- [clarifications and additional information \(https://www.gov.uk/guidance/recyclability-assessment-methodology-supplementary-guidance\)](https://www.gov.uk/guidance/recyclability-assessment-methodology-supplementary-guidance) to help you carry out your RAM assessment
- [the RAM decision tree \(https://www.gov.uk/government/publications/recyclability-assessment-methodology-decision-tree-diagram\)](https://www.gov.uk/government/publications/recyclability-assessment-methodology-decision-tree-diagram)

Stages of recyclability

This methodology is focused on evaluating recyclability, encompassing the full range of requirements across the packaging end-of-life stages. It assesses whether packaging can be recycled and transformed into new product applications - that is, replacing virgin materials. By considering factors such as collection, sortation, and reprocessing capabilities, the methodology aims to support a circular packaging economy, where materials are recaptured and recycled rather than disposed of, while also addressing the practical challenges faced across the value chain.

There are 5 end-of-life stages providing the foundational structure for the methodology.

1. Classification

This defines how to assess whole packaging units and components against the RAM.

Multi-material packaging can be constructed of assembled components of different materials. Where these components can be easily separated by hand, each component is to be assessed separately. Packaging items that are separated during product use, and are not able to be reassembled, must also be assessed separately (for example tear off strip, wine bottle cork).

Multi-material packaging of assembled components of different materials which cannot be easily separated by hand or where one component is made of combination of materials should be assessed under the predominant material by weight. But each material must be reported according to the Regulations, using the **red** / **amber** / **green** output assigned to the combined unit. output assigned to the combined unit. If the different materials are of equal weight, then the materials must be assessed separately under the relevant material categories.

Where the material or a whole unit of packaging is not able to be defined as one of the material categories, that component is to be assessed within the '[Other materials](https://www.gov.uk/government/publications/recycling-assessment-methodology-materials-and-outputs/recycling-assessment-methodology-assessing-materials#other-materials) (<https://www.gov.uk/government/publications/recycling-assessment-methodology-materials-and-outputs/recycling-assessment-methodology-assessing-materials#other-materials>)' packaging category.

Example: jam jar

For a jar of jam comprising a glass jar, a paper label, a steel lid, and a hessian lid cover and tie should be assessed and reported as:

	Jar	Label	Lid	Tie	Hessian Lid
Material category assessed in RAM	Glass	Glass (because of predominant material)	Steel	Other (assumed textile)	Other (assumed textile)
RAM output (illustrative only)	green	green	green	red	red
Reported as on RPD	Glass - green	Paper and board - green	Steel - green	Other - red	Other - red

Where the label and jar are not easily separable by hand they are assessed collectively in the RAM, applying the predominant material rule to determine the material category. However, each material must be reported in accordance with the regulations, along with the assigned RAM output (**Red**, **Amber** or **Green**). This would mean that the label and jar are reported separately in RPD, but with a common **green** RAM output.

Example: pack of batteries

A pack of 4 batteries in packaging made from a clear moulded plastic cover that is glued to a cardboard backing, where the plastic can be separated from the card by hand must be assessed and reported separately as:

Cover Backing

	Cover	Backing
Material category assessed in RAM	Plastic - rigid	Paper and board
RAM output (illustrative only)	amber	green
Reported as on RPD	Plastic - amber	Paper and board - green

Example: bottle of detergent

A detergent bottle made from a clear plastic PET bottle, PP cap and PE label should be assessed as a whole unit made of the same material as the predominant component.

	Detergent bottle	Cap	Label
Material category assessed in RAM	Plastic - rigid (predominant material)	Plastic - rigid (undersize for separate assessment)	Plastic - rigid
RAM output (illustrative only)	green	green	green
Reported as on RPD	Plastic - rigid - green	Plastic - rigid - green	Plastic - flexible - green

Where the cap is an undersized item not meeting the dimension threshold for sortation it can be attached to another component – bottle – and should be considered as one component to continue the assessment. There is more on undersized components in the [guidance for plastics - rigid \(https://www.gov.uk/government/publications/recycling-assessment-methodology-materials-and-outputs/recycling-assessment-methodology-assessing-materials#plastic-rigids\)](https://www.gov.uk/government/publications/recycling-assessment-methodology-materials-and-outputs/recycling-assessment-methodology-assessing-materials#plastic-rigids).

However, each material must be reported in accordance with the regulations along with the assigned RAM output (e.g. **red**, **amber**, **green**). This would

mean that the bottle, label and lid are reported separately in RPD, but with a common **green** RAM output.

Example: liquid drink carton

A liquid drink carton (not meeting the definition of Drink Container) with the carton made of a fibre-based composite material with a plastic cap.

	Carton	Cap
Material category assessed in RAM	Fibre-based composite (predominant material)	Fibre-based composite (undersize for separate assessment)
RAM output (illustrative only)	amber	amber
Reported as on RPD	Fibre-based composite - amber	Plastic - rigid - amber

Where the cap is an undersized item not meeting the dimension threshold for sortation it can be attached to another component – bottle – and should be considered as one component to continue the assessment. There is more on undersized components in the [guidance for fibre-based composites \(https://www.gov.uk/government/publications/recycling-assessment-methodology-materials-and-outputs/recycling-assessment-methodology-assessing-materials#fibre-based-composite-materials\)](https://www.gov.uk/government/publications/recycling-assessment-methodology-materials-and-outputs/recycling-assessment-methodology-assessing-materials#fibre-based-composite-materials).

However, each material must be reported in accordance with the regulations along with the assigned RAM output (e.g. **red**, **amber**, **green**). This would mean that the carton and lid are reported separately.

2. Collection

This is the main route to recycling for a given product and refers to one of the three options below. More detail of what is and is not accepted for each material type is included within the guidance in section 3. There are three

main routes to recyclability.

Widely collected (via kerbside)

Item of packaging or component must be collected at kerbside by at least 75% of local authorities (LAs) across the UK, as determined by WRAP's local authority data.

Limited collection (via kerbside)

Some items of packaging and components are collected by a substantial number of local authorities across the UK but fall short of meeting the 75% threshold for widely accessible collection at kerbside. These items may proceed via the limited collection route if collected at kerbside by at least 50% of local authorities (LAs) across the UK, as determined by WRAP's local authority data.

Take-back schemes

Dedicated take-back schemes offer a way to recapture materials and packaging types that are not included in kerbside collections. Obligated producers must confirm their chosen scheme meets the following criteria:

- accessible to at least 75% of the UK population
- does not conflict with local authority kerbside collections
- is not restricted to one brand or product
- does not require a purchase to be made before an item can be deposited
- have full transparency that the collected material is recycled

There is a full description of these factors in the section on [standards for take-back protocols](#).

3. Sortation

This refers to the process of capturing and diverting packaging waste into appropriate waste streams for further reprocessing. In some local authorities, this separation occurs more extensively at the kerbside by consumers (for example, source separated versus comingled collection systems). However, for the purposes of this methodology, the focus is on the ability to sort waste at scale within materials facilities (MFs). Packaging specifications and design choices, such as material shape, size, density, and colour, play a key role in ensuring MFs can efficiently identify and sort packaging waste. While some packaging items may technically be sortable, various factors can hinder their effective separation in sortation facilities. Future iterations of the RAM will seek to leverage MF sampling data to provide a more accurate assessment of the sortation capabilities and efficiencies for each packaging material and format.

4. Reprocessing

This is the technical process of creating recyclate from each of the packaging waste streams. Each material type requires specific reprocessing technologies, and contaminants can disrupt the process or lower the quality of the recycled materials. The tolerance for contaminants varies by material and process, with different items and substances prohibited in each reprocessing method.

5. Application

This refers to factors which may affect the quality of recyclate produced or cause unnecessary secondary material loss in the recycling process. Recycling is defined in the regulations as any recovery operation by which waste materials are reprocessed into products, materials, or substances, whether for their original purpose or other purpose. It does not include energy recovery and the reprocessing into materials that are to be used as fuels or for backfilling operations.

Standards for take-back protocols

This protocol is aligned to the [OPRL \(https://oprl.org.uk/\)](https://oprl.org.uk/) Takeback Protocol to ensure consistency with existing industry labelling standards and practices.

Accessibility

Collection points must be available to at least 75% of the population. This must be proven by a robust methodology that accounts for travel distance and the range of transport options available, for example 'drive time' to the collection point. Collection points must also be clearly signposted and inclusive of those with disabilities. They should be accessible throughout normal business hours and within a maximum 5 mile radius for 75% or more of the UK population.

To avoid double counting populations in overlapping geographic areas, it's important to implement a deduplication method for the catchment areas surrounding each collection point. If a geographic area falls within the drivetime catchments of multiple collection points, the most frequently used collection point should be selected to prevent population overlap.

No conflict with kerbside collections

Collection schemes should not actively compete with kerbside collections or encourage consumers to use their system in place of kerbside collections.

No brand restrictions

Schemes must accept packaging from other brands that have a similar format.

No purchase necessary

Customers cannot be required to provide proof or purchase to use the service.

Traceability

The scheme must provide a defined end market for the reprocessed material and evidence must be made available upon request that confirms the tons of material reprocessed. Contaminating materials must also be included in this data request.

Definitions

Definitions are aligned to The Producer Responsibility Obligations (Packaging and Packaging Waste) Regulations 2024. Where additional definitions are given, references are provided.

Alternative natural fibres

For the purposes of this assessment this term refers to alternatives to cellulosic wood-based fibres such as bagasse, sugar cane, palm, fibre, rice straw, wheat straw, barley straw, oat straw, grass straw, flax, hemp, and bamboo^{[\[footnote 1\]](#)}

Board/paperboard

A generic term applied to certain types of paper frequently characterised by their relative high rigidity. The primary distinction between paper and board

is normally based upon thickness or grammage, though in some instances the distinction will be based on the characteristics and/or end-use.

Component

The different parts of a whole packaging unit that are separable by hand.

Drink container

Drink container means a bottle or can which:

- contains or used to contain drink;
- is made wholly or mainly from polyethylene terephthalate (PET) plastic, glass, steel or aluminium;
- has a capacity of at least 150 millilitres but no more than 3 litres of liquid;
- is designed or intended to be sealed in an airtight and watertight state at the point of supply to a consumer in the United Kingdom; and
- is not conceived, designed or marketed to be refilled or reused in any other way by any person

Fibre-based composite packaging

This is made of paperboard or paper fibres, with one or more layers of plastic, and which may also have layers of other materials, to form a single unit that cannot be separated by hand.

If the producer who supplies the packaging can provide evidence that its plastic content is not more than 5% of the packaging material by mass, then the material can be classified as 'paper and board' instead.

Liable producer

A producer who is liable under regulation 60 of the Producer Responsibility Obligations (Packaging and Packaging Waste) Regulations 2024, to pay a disposal fee and an administration fee to the scheme administrator for that assessment year.

Non-paper content

The sum of all non-separable components of a fibre-based packaging product that are not defined as paper content expressed in percentage of total component weight. This means that all films, foils, coatings (except coatings mentioned in paper content definition), inks, varnishes would be excluded from the paper content.

Packaging materials

Materials used in the manufacture of packaging and includes raw materials and processed materials prior to the conversion into packaging.

Paper

Paper consists mainly of natural cellulosic fibres and may contain other ingredients such as filling material, starch, clay or colour coatings including binder, as well as additives typically used in the paper industry such as wet-strength agents, sizing agents, dyes and bound water.

Paper content

The sum of all non-separable components of a fibre-based packaging product that are defined as paper content expressed in percentage of total component weight. The paper content includes natural cellulosic wood fibres, paper making additives (e.g. wet strength, sizing agents, fillers and dyes), water content and clay coatings or colour coatings.

Plastics (films/flexibles)

Plastics classified as 'flexible' refer to packaging structures which flex easily, and which can be made of both single and multiple layers of materials. Common examples of flexible plastic packaging include bread bags, snack wrappers, netted produce bags, zipper-lock pouches, lidding film and sachets.

Plastics (rigids)

Plastics classified as 'rigid' in packaging refer to items that maintain a defined shape and structural integrity under normal conditions of use. These materials are typically stiff and durable, offering protection and support for a wide range of products. Common examples of rigid plastics in packaging include bottles, pots, tubs, trays, tubes, caps, and closures.

Predominant material

Where a whole unit of packaging or a packaging component is comprised of two or more materials, the predominant material is the one that is greatest by weight.

Primary packaging

Packaging conceived so as to constitute a sales unit to the final user or consumer at the point of purchase.

Printing inks

Printing inks are mixtures of colourants with other substances which are applied on materials to form a graphic or decorative design. They can be applied together with or without other coloured or uncoloured overprint varnishes or coatings, or primers, which are normally applied in combination with the colourants in order to enable the printed design to achieve specific functions such as ink adhesion, rub resistance, gloss, slip or friction, and durability.

Printing inks do not include coatings which are applied with the prime objective of enabling the material or article to achieve a technical function such as heat sealing, barrier, or corrosion resistance, as opposed to a graphic effect, even though they may be coloured.

Recycling

Has the meaning given in Article 3(17) of the Waste Framework Directive: “Any recovery operation by which waste materials are reprocessed into products, materials or substances whether for the original or other purposes. It includes the reprocessing of organic material but does not include energy recovery and the reprocessing into materials that are to be used as fuels or for backfilling operations.”[\[footnote 2\]](#)

Regulator

This is the ‘appropriate agency’ in the Producer Responsibility Obligations (Packaging and Packaging Waste) Regulations 2024. In England this is the Environment Agency. In Wales this is the Natural Resources Body for Wales (NRW). In Scotland this is the Scottish Environment Protection Agency (SEPA). In Northern Ireland this is the Department of Agriculture, Environment and Rural Affairs for Northern Ireland (DAERA).

Re-use

Any operation by which products or components that are not waste are used again for the same purpose for which they were conceived.

Secondary packaging

This is packaging that:

- is conceived so as to constitute at the point of purchase a grouping of a certain number of sales units, whether that grouping is supplied as such to the final user or consumer or whether it serves only as a means to replenish the shelves at the point of sale
- can be removed from the product without affecting the product's characteristics

Shipment packaging

Refers to tertiary packaging on, or that is used to contain, items which are supplied or are intended to be supplied to a consumer.

Tertiary packaging

Is conceived so as to facilitate the handling and transport of one or more sales units, or secondary packaging, for example by preventing damage from physical handling and transport damage; and is not a road, rail, ship or air container

Unit

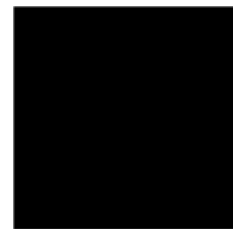
A whole unit of packaging can be made up of one or multiple components that are combined together to create the complete item of packaging.

1. [CPI Design for Recyclability Guidelines 4th Edition \(https://paper.org.uk/CPI/CPI/Content/Information/Publications.aspx\)](https://paper.org.uk/CPI/CPI/Content/Information/Publications.aspx)
2. [Article 3\(17\) of the Waste Directive](#)



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